The data verbs: select · filter · mutate · arrange

Tuesday, June 2, 2026

Today: Tuesday June 2

Before class

- **Perusall Assignment**: *DC* §10.1–10.6

Plan for today

- Four new data verbs: `filter`, `arrange`, `select`, `mutate`
- Sorting verbs by *what they touch*: rows vs. columns
- Combining verbs with the pipe into real pipelines
- Formatting pipelines so they stay readable

Helpful links

- Syllabus · AI policy · Schedule · *R4DS* Ch. 3 · *R4DS* Ch. 4 · *DC* Ch. 10 · dplyr reference

Recap from yesterday

We discussed the overarching **framework** and two data verbs:

Family	Input to Output	The verb that uses it
Reduction	variable to scalar	`summarise()`
Transformation	variable to variable	*…today:* ***`mutate()`***
Data verb	data frame to data frame	all of them

We write reduction functions inside of `summarise()`. Today we will cover `mutate()`, which contains **transformation** functions, plus three verbs that reshape *which* rows and columns you keep.

A map of the verbs

We can organize dplyr by **what each verb operates on**:

Table 1: The dplyr toolkit.

Touches…	Verbs	What changes
Rows	`filter()`, `arrange()`	which rows, and their order
Columns	`select()`, `mutate()`	which columns, and new ones
Groups	`group_by()`, `summarise()`	*(yesterday)*

> Tip
>
> These all follow the same three rules: **first argument is a data frame, you name columns without quotes, and the output is a brand-new data frame.**

Rows: filter()

filter(): keep the rows you want

`filter()` keeps only the rows that pass a test (*DC* §10.2, R4DS §3.2.1):

```
penguins |>
  filter(species == "Gentoo")
```

```
# A tibble: 124 x 8
   species island bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
   <fct>   <fct>           <dbl>         <dbl>             <int>       <int>
 1 Gentoo  Biscoe           46.1          13.2               211        4500
 2 Gentoo  Biscoe           50            16.3               230        5700
 3 Gentoo  Biscoe           48.7          14.1               210        4450
 4 Gentoo  Biscoe           50            15.2               218        5700
 5 Gentoo  Biscoe           47.6          14.5               215        5400
 6 Gentoo  Biscoe           46.5          13.5               210        4550
```

```
 7 Gentoo  Biscoe           45.4          14.6               211        4800
 8 Gentoo  Biscoe           46.7          15.3               219        5200
 9 Gentoo  Biscoe           43.3          13.4               209        4400
10 Gentoo  Biscoe           46.8          15.4               215        5150
# i 114 more rows
# i 2 more variables: sex <fct>, year <int>
```

The test is a **transformation** that returns `TRUE`/`FALSE` for each row; `filter()` keeps the `TRUE`s.

The comparison operators

These are the tests you'll use inside `filter()` (*DC* Table 10.9):

Table 2: Comparisons for `filter()`.

Operator	Means	Example
`==`	equal to	`species == "Gentoo"`
`!=`	not equal to	`sex != "male"`
`> >=`	greater (or equal)	`body_mass_g >= 5000`
`< <=`	less (or equal)	`flipper_length_mm < 190`
`%in%`	is one of	`island %in% c("Dream", "Biscoe")`

> **Warning**
>
> Use `==` to **test** equality, not `=`. A single `=` means "assign," not "test equality".

Multiple conditions

Separate tests with a comma (or `&`) for **AND**, a vertical bar `|` for **OR** (R4DS §3.2.1):

```r
# AND: Adelies that are ALSO on Dream
penguins |>
  filter(species == "Adelie", island == "Dream")

# OR: any penguin that is Adelie OR Gentoo
penguins |>
  filter(species == "Adelie" | species == "Gentoo")
```

```r
# %in% is the clean way to write a long OR
penguins |>
  filter(species %in% c("Adelie", "Gentoo"))
```

> **Note**
>
> `filter(species == "Adelie" | "Gentoo")` does **not** work the way it reads. Each side of `|` must be a complete test.

Dropping `NA`s

Yesterday, `group_by(species, sex)` gave us an `NA` group because some penguins have a missing `sex`. Now we can drop them:

```r
penguins |>
  filter(!is.na(sex)) |>
  group_by(species, sex) |>
  summarise(n = n(), .groups = "drop")
```

```
# A tibble: 6 x 3
  species   sex        n
  <fct>     <fct>  <int>
1 Adelie    female    73
2 Adelie    male      73
3 Chinstrap female    34
4 Chinstrap male      34
5 Gentoo    female    58
6 Gentoo    male      61
```

`is.na(sex)` is `TRUE` for the missing ones; `!` flips it, so `!is.na(sex)` keeps the rows that **do** have a sex. This removes the `NA` group.

Rows: arrange()

arrange(): set the order

`arrange()` reorders rows without adding or removing any. This function gives the result in ascending order by default:

```
penguins |>
  arrange(body_mass_g)
```

```
# A tibble: 344 x 8
   species   island   bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
   <fct>     <fct>             <dbl>         <dbl>             <int>       <int>
 1 Chinstrap Dream              46.9          16.6               192        2700
 2 Adelie    Biscoe             36.5          16.6               181        2850
 3 Adelie    Biscoe             36.4          17.1               184        2850
 4 Adelie    Biscoe             34.5          18.1               187        2900
 5 Adelie    Dream              33.1          16.1               178        2900
 6 Adelie    Torgers~           38.6          17                 188        2900
 7 Chinstrap Dream              43.2          16.6               187        2900
 8 Adelie    Biscoe             37.9          18.6               193        2925
 9 Adelie    Dream              37.5          18.9               179        2975
10 Adelie    Dream              37            16.9               185        3000
# i 334 more rows
# i 2 more variables: sex <fct>, year <int>
```

You can wrap a column in `desc()` if you want descending order. If there are ties, to break them you can list several columns:

```
penguins |>
  arrange(species, desc(body_mass_g))   # by species, then heaviest first
```

Columns: select()

select(): keep the columns you want

`select()` is to columns what `filter()` is to rows:

```
penguins |>
  select(species, island, body_mass_g)
```

```
# A tibble: 344 x 3
   species island    body_mass_g
   <fct>   <fct>           <int>
 1 Adelie  Torgersen        3750
 2 Adelie  Torgersen        3800
```

```
 3 Adelie  Torgersen        3250
 4 Adelie  Torgersen          NA
 5 Adelie  Torgersen        3450
 6 Adelie  Torgersen        3650
 7 Adelie  Torgersen        3625
 8 Adelie  Torgersen        4675
 9 Adelie  Torgersen        3475
10 Adelie  Torgersen        4250
# i 334 more rows
```

You can grab a range, drop with `!`, or rename as you go:

```
penguins |> select(species:bill_depth_mm)  # a range of columns
penguins |> select(!year)                  # everything EXCEPT year
penguins |> select(mass = body_mass_g)     # rename while selecting
```

> **Note**
>
> `penguins` has only 8 columns, so `select()` feels optional here. On a 19-column table like `flights` from HW2, or a 76-column one, it's one of the first things you might want to try.

select() helpers

For wide tables, name columns by pattern instead of one-by-one:

```
penguins |> select(starts_with("bill"))    # bill_length_mm, bill_depth_mm
penguins |> select(ends_with("_mm"))       # all the millimetre measurements
penguins |> select(where(is.numeric))      # every numeric column
```

Columns: mutate()

mutate(): add a new column

`mutate()` adds columns computed from existing ones, keeping every row:

```
penguins |>
  mutate(body_mass_kg = body_mass_g / 1000) |>
  select(species, body_mass_g, body_mass_kg)
```

```
# A tibble: 344 x 3
   species body_mass_g body_mass_kg
   <fct>         <int>        <dbl>
 1 Adelie         3750         3.75
 2 Adelie         3800         3.8
 3 Adelie         3250         3.25
 4 Adelie           NA        NA
 5 Adelie         3450         3.45
 6 Adelie         3650         3.65
 7 Adelie         3625         3.62
 8 Adelie         4675         4.68
 9 Adelie         3475         3.48
10 Adelie         4250         4.25
# i 334 more rows
```

mutate() contains transformation functions

> `mutate()` **always** involves a *transformation* function, just as `summarise()` **always** involves a *reduction* function.

```
# transformation: one value per row to mutate()
penguins |> mutate(bill_ratio = bill_length_mm / bill_depth_mm)

# reduction: one value total to summarise()
penguins |> summarise(avg_bill = mean(bill_length_mm, na.rm = TRUE))
```

> **Tip**
>
> Deciding between them is the same question from yesterday: does the calculation give **one value per row** (`mutate`) or **one value for the whole group** (`summarise`)?

A categorical column with ifelse()

`ifelse()` is a transformation that picks between two values per row, perfect inside `mutate()`:

```
penguins |>
  mutate(size = ifelse(body_mass_g > 4000, "large", "small")) |>
  select(species, body_mass_g, size)
```

```
# A tibble: 344 x 3
   species body_mass_g size
   <fct>         <int> <chr>
 1 Adelie         3750 small
 2 Adelie         3800 small
 3 Adelie         3250 small
 4 Adelie           NA <NA>
 5 Adelie         3450 small
 6 Adelie         3650 small
 7 Adelie         3625 small
 8 Adelie         4675 large
 9 Adelie         3475 small
10 Adelie         4250 large
# i 334 more rows
```

Combining verbs

The pipe does the real work

One verb does one thing. For most questions, we will need several, chained together with `|>`:

```
penguins |>
  filter(!is.na(sex)) |>
  mutate(body_mass_kg = body_mass_g / 1000) |>
  select(species, sex, body_mass_kg) |>
  arrange(desc(body_mass_kg))
```

This can be read in English as: take penguins, **then** drop missing sex, **then** add mass in kg, **then** keep three columns, **then** sort heaviest first.

> Tip
>
> Because each verb returns a new data frame, the output of one is the input to the next. The verbs line up on the left so you can quickly skim to understand what's going on.

Putting all six together

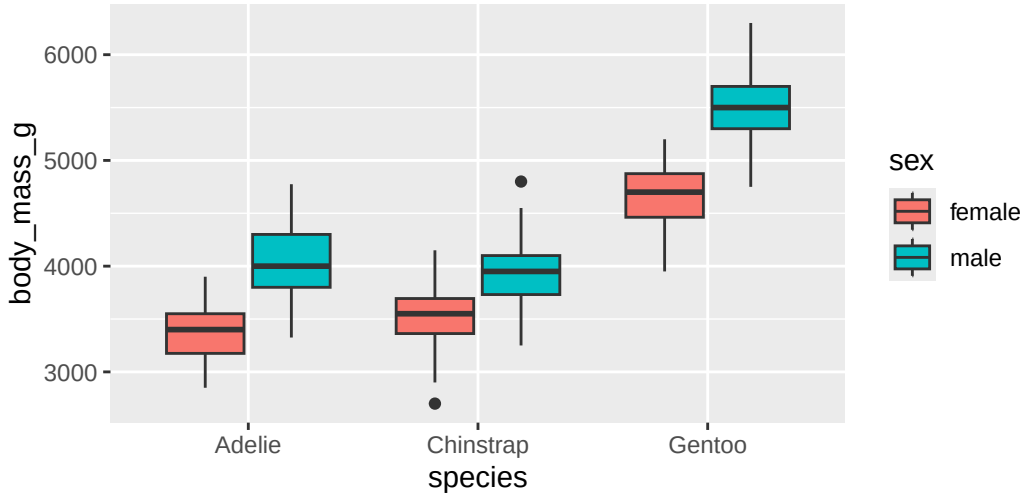
We can use all six verbs to answer the question, "what's the average mass of each species/sex, biggest first?":

```r
penguins |>
  filter(!is.na(sex)) |>
  group_by(species, sex) |>
  summarise(
    n        = n(),
    avg_mass = mean(body_mass_g, na.rm = TRUE),
    .groups  = "drop"
  ) |>
  arrange(desc(avg_mass))
```

```
# A tibble: 6 x 4
  species   sex        n avg_mass
  <fct>     <fct>  <int>    <dbl>
1 Gentoo    male      61    5485.
2 Gentoo    female    58    4680.
3 Adelie    male      73    4043.
4 Chinstrap male      34    3939.
5 Chinstrap female    34    3527.
6 Adelie    female    73    3369.
```

And because the result is just a data frame, you can pipe it straight into a plot (switching `|>` to `+`):

```r
penguins |>
  filter(!is.na(sex)) |>
  ggplot(aes(x = species, y = body_mass_g, fill = sex)) +
  geom_boxplot()
```



Formatting your pipelines

A multi-step pipeline is only readable if you format it. The rules that matter most:

- A space before `|>`, and let `|>` be the **last thing on the line**.
- For verbs with **named arguments** (`mutate`, `summarise`), put each argument on its **own line**, indented two spaces.
- Verbs without named arguments (`filter`, `select`, `arrange`) can stay on one line if they fit.
- Name new columns in `snake_case`: `body_mass_kg`, not `BodyMassKG`.

```
# Strive for                          # Avoid
penguins |>                           penguins|>filter(!is.na(sex))|>group_by(
  filter(!is.na(sex)) |>              species)|>summarize(n=n(),avg=mean(
  group_by(species) |>                body_mass_g,na.rm=TRUE))
  summarise(
    n   = n(),
    avg = mean(body_mass_g, na.rm = TRUE)
  )
```

> **Tip**
>
> Don't hand-format forever: the **`styler`** package reformats a whole file for you. In RStudio, `Cmd/Ctrl + Shift + P` then type "styler". Full guide: R4DS Ch. 4.

A helpful shortcut

Counting rows per group is so common it has its own verb, `count()`:

```
penguins |> count(species)

# is exactly the same as
penguins |>
  group_by(species) |>
  summarise(n = n())
```

> **Note**
>
> `count()` is just `group_by()` + `summarise(n = n())` bundled up. Add `sort = TRUE` to get the biggest groups first.

What we covered today

- `filter()`: keep rows that pass a test; `==` not `=`; combine with `,`/`|`/`%in%`; `!is.na()` to drop missing
- `arrange()`: reorder rows; `desc()` for descending; extra columns break ties
- `select()`: keep/drop/rename columns; `starts_with()` & friends for wide tables
- `mutate()`: add columns from transformations; the per-row twin of `summarise()`
- **The pipe**: chain verbs into a readable top-to-bottom story; pipe results into ggplot
- **Style**: space before `|>`, named args on their own lines, `snake_case`, let `styler` do the rest

Activity

Activity: wrangling the penguins

Roughly 25 minutes. Individually or with a neighbor.

- Activity Canvas Link

Format: a handout that walks each verb, then asks you to chain them. You'll match tasks to verbs, write `filter`/`arrange`/`select`/`mutate` pipelines on `penguins`, debug a `=` vs `==` error, build one multi-verb pipeline that answers a question, and restyle a deliberately unreadable pipeline.

To turn in: save your `.qmd` and rendered HTML.

Wrap-up & looking ahead

Tomorrow (Wednesday): no new reading, we put all six verbs together on a messier dataset and turn raw rows into summary tables and plots.

Upcoming Deadlines

- **Thu 6/4, before class**: *DC* §12.1
- **Thu 6/4, 11:59 p.m.**: Project: Team formation
- **Fri 6/5, 11:59 p.m.**: Homework 3: Data Wrangling & Reshaping

Reach out

Stuck? Office hours, or email me (cgc5478@psu.edu) or Xinyue (xpw5228@psu.edu).

Reference

Syllabus · AI policy · Schedule · *R4DS* Ch. 3 · *R4DS* Ch. 4 · *DC* Ch. 10 · dplyr reference